Radiation Reaction in Ultra-High Magnetic Fields and Highest Energy Cosmic Rays with Their Neutrinos

acceleration

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Electromagnetic radiation damping reaction is prominent in strong electric and magnetic fields. The Lorentz-Dirac equation (LDE) is the equation to describe the charged particle motion under the external force as well as its own electromagnetic field in the framework of classical relativistic electrodynamics [1]. The problems of LDE are that, this second order differential equation contains unphysical solutions such as the runaway, preacceleration and ununiqueness solutions [1, 2]. Also, one can find the released electromagnetic energy from a particle exceeds its total energy γmc^2 with mass m and Lorentz factor γ due to an extra damping term proportional to $d(d\gamma/d\tau)/dt$ [3]. Consequently, a modified LDE is proposed to eliminate the problems, and also balance the particle mechanism energy with electromagnetic energy [2, 3],

$$ma^\alpha = F^\alpha_{\rm ext} + b(\delta^\alpha_\beta + c^{-2}u^\alpha u_\beta)\dot{F}^\beta_{\rm ext}, \tag{1}$$

where $F_{\rm ext}$ is the external force, and $b = 2e^2/(3mc^3) \sim r_0/c$ is the so-called reaction damping timescale with $r_0 = e^2/(mc^2)$ as the classical particle radius. This modified equation is valid only if the external force varies slowly enough to cover the classical size of the particle r_0, otherwise the particle should be treated as extended source rather than a point.

Radiation damping reaction is important when the magnetic field is extremely high. Previous work usually adopted radiation reaction as the perturbation term. In this work we will show an analytic solution of charged particle motion in strong fields. We apply it to particle acceleration, and show the principally maximum energy of ultra-high energy cosmic rays (UHECRs) in the astrophysical environment. Generally speaking UHECRs can be accelerated via shock acceleration by the first or second order of Fermi mechanism [4], up to energies $10^{20} - 10^{21}$ eV [5]. Recently Thompson and Lacki showed that in principle protons are possible to be accelerated to energies around $\sim 10^{24}$ eV according to the maximum possible luminosity [6]. On the other hand, electromagnetic fields of compact objects also accelerate UHECRs [7–12], but it is debatable whether radiation loss is significant constrain the particle highest energy [8, 12]. In this paper we show that, charged particles can be accelerated and achieve an energy of $\sim 3\times 10^{26}$ eV around compact object surface with very strong electromagnetic fields, as no cooling mechanisms can dominate over electric field acceleration. The following produced secondary cosmic rays (???) and neutrinos are predicted for future detection.

Equations. – Initially we give a time-independent electric field $\mathbf{E}_l = E_l(0,1,0)$ and magnetic field $\mathbf{B} = B(0,0,1)$, which are orthogonal with each other and $E_l = B$. The local Cartesian coordinate is set along the tangents of the two fields. Here the footnote l in the electric field is to distinguish it with particle energy symbol E. The four components of the modified LDE is written as the differential equations of proper time τ as

$$\begin{aligned}
\ddot{\xi}_1 &= \omega_0\dot{\xi}_2 + b\omega_0\ddot{\xi}_2 + b\omega_0 c^{-2}\dot{\xi}_1(\dot{\xi}_1\ddot{\xi}_2 - \dot{\xi}_2\ddot{\xi}_1),\\
\ddot{\xi}_2 &= -\omega_0\dot{\xi}_1 - b\omega_0\ddot{\xi}_1 + b\omega_0 c^{-2}\dot{\xi}_2(\dot{\xi}_1\ddot{\xi}_2 - \dot{\xi}_2\ddot{\xi}_1) + \frac{eE_l\gamma}{m},\\
\ddot{\xi}_3 &= b\omega_0 c^{-2}\dot{\xi}_3(\dot{\xi}_1\ddot{\xi}_2 - \dot{\xi}_2\ddot{\xi}_1)\\
\ddot{t} &= b\omega_0 c^{-2}\dot{t}(\dot{\xi}_1\ddot{\xi}_2 - \dot{\xi}_2\ddot{\xi}_1) + \frac{e}{mc^2}E_l\dot{\xi}_2\gamma^2,
\end{aligned} \tag{2}$$

where $\omega_0 = eB/(mc)$, and γ is the Lorentz factor of the charge particle in the comoving frame. From equation (2) we obtain $d^2\xi_3/dt^2 = -(d\xi_3/dt)[eE_l/(me\gamma)]$, which means $d\xi_3/dt$ is always decreasing along the $z-$axis, even there is no external force in that direction. For simplicity, we consider a two-dimensional motion perpendicular to the magnetic field. Then it is easier to solve equation (2) by taking $\dot{\xi}_1(\dot{\xi}_2) = c\sinh q\cos\theta(\sin\theta)$ and $\dot{t} = c\cosh q$ [13]. The analytic solution can be derived when we adopt an average angle $\theta \approx \theta_0$ in the electric field term of the equation, which is reasonable when the involved timescale is much less than the variation of θ. The solution gives the electromagnetic field energy total lost $P_{\rm em} = 2e^2 a_\mu a^\mu/(3c^3)$ as

$$P_{\rm em} = \frac{2e^2}{3c}\omega_0^2\frac{\gamma^2(1-\cos\theta_0)^2 + \sin^2\theta_0}{1 + b^2\omega_0^2\gamma^2}, \tag{3}$$

and the curvature radius as

$$R_c = \left(\frac{E}{eB}\right)\frac{1 + b^2\omega_0^2\gamma^2}{|(1-\cos\theta_0) + b\omega_0\sin\theta_0|}, \tag{4}$$

where the particle total energy $E = \gamma c^2$. We consider the extreme case that $b\omega_0 > 1$, when charged particles shows very different behavior from the ordinary case $b\omega_0 \ll 1$. Now the electromagnetic emitted power is independent of γ as a constant $P_{\rm em} \simeq 3m^2c^5/[2e^2(1-\cos\theta_0)^2]$, which is very different from the classical perpendicular acceleration $P \propto \gamma^2$ (synchrotron radiation) or parallel acceleration $P_{\rm em} \propto \gamma^4$ (curvature radiation). Also, the trajectory of the particle goes as a straight for large γ. The total electromagnetic radiation timescale is $t_{\rm em} = E/P_{\rm em} \simeq 2e^2\gamma/(mc^3) \simeq b\gamma$. One can check that the timescale of θ variation $t_\theta = 2\pi/|d\theta/dt| \simeq 2\pi b\gamma^3 \gg t_{\rm em}$, which shows for a single particle the above average $\theta \simeq \theta_0$ is reasonable if we focus on a timescale less than $t_{\rm em}$. Also, the variation of the external force can be measured via the

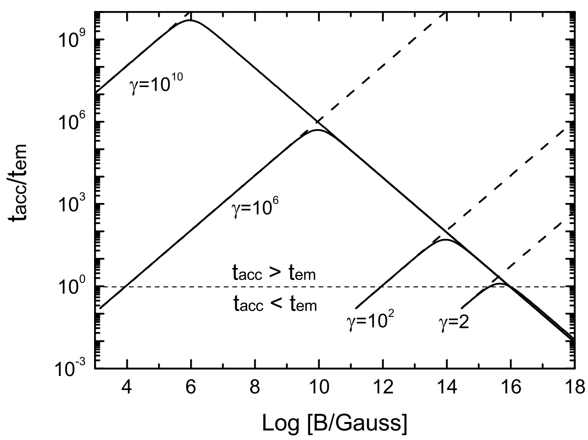

FIG. 1: Comparison between electron acceleration and electromagnetic cooling timescale $t_{\rm acc}/t_{\rm em}$ as a function of magnetic field. The solid lines gives $t_{\rm acc}/t_{\rm em}$ including radiation reaction, but dashed lines without radiation reaction. The particle Lorentz factor is adopted as $\gamma = 2, 10^2, 10^6, 10^{10}$. Plots are constrained in the region $t_{\rm em} < t_\theta$ where the angle θ is taken as $\pi/2$.

ratio $|b\dot{F_{\rm ext}}/F_{\rm ext}|$ in the particle rest frame. We obtain $|b\dot{F_{\rm ext}}/F_{\rm ext}| \to b\omega_0(1+b^2\omega_0^2)^{-1/2} < 1$ at $\gamma \to 1$ for the rest particle. This means the external force variation always covers the entire classical particle, which can be treated as a point source. The limit case is that the external force variation timescale is just reaching the damping timescale b for $b\omega_0 \to \infty$. We take it as the critical case that the modified LD equation can still be valid.

Next we go to particle acceleration. The particle acceleration timescale in the comoving frame by electric field is $t_{\rm acc} = (\gamma mc)/(eE\sin\theta_0) = \gamma mc/(eB\sin\theta_0)$. The necessary requirement for electric field acceleration is the acceleration time being less than the electromagnetic cooling time $t_{\rm acc} < t_{\rm em}$, which gives $B > [3e(1-\cos\theta_0)^2]/(r_0^2\sin\theta_0)$. We take B to be larger than the maximum of the right side (along the y-axis at $\theta_0 \simeq 0$), which is just consistent with the above assumption $b\omega_0 > 1$. Thus we have the criterion for electric field acceleration as $B > 3e/(2r_0^2)$. For electrons, this criterion gives $B > B_{\rm cirt} = 9\times 10^{15}(m/m_e)^2$ G. The ratio $t_{\rm acc}/t_{\rm em}$ is showed in Fig. 1 as the function of magnetic fields. Radiation reaction decreases the acceleration timescale dramatically in strong field at $B\gamma \geq 3e/(2r_0^2)$ and becomes $t_{\rm acc} < t_{\rm em}$ at $B_{\rm cirt}$ no matter the initial γ.

More generally, we introduce a parameter $\epsilon = E_l/B$ and consider a shell which is ejected into the space along $z-$axis with a bulk Lorentz factor Γ (velocity β). In the shell comoving frame the electric and magnetic fields are $\mathbf{E}_l = E_l(0,\Gamma,0)$ and $\mathbf{B}' = B(\beta\Gamma,0,1)$. Using the similar method we get the criterion for particle acceleration in the shell as

$$B' = B\Gamma > 3e/(2\epsilon' r_0^2) \simeq 9\times 10^{15}\epsilon'^{-1}(m/m_e)^2\,\mathrm{G} \quad (5)$$

where $\epsilon' = [\beta^2 + (\epsilon\Gamma)^{-2}]^{-1/2}$.

Electron acceleration. – Now we consider the electromagnetic field comes from an astrophysical compact object with mass M, surface magnetic field B_* and radius r_*. (It is also possible that the magnetized torus around the object also contributes to the total fields). The electric field is perpendicular to the magnetic field, and we take $\epsilon = E_l/B_*$ near the surface. The electromagnetic field can be seen as steady-state if $b \ll r_*/c$, which can be satisfied almost everywhere. One previously ignored acceleration mechanism of UHECRs, which can be achieved due to radiation reaction, is the electromagnetic field acceleration near the object surface. According to equation (5), acceleration can dominate over radiation loss if the magnetic field satisfies $B_* > 3e/(2\epsilon r_0^2)$. The maximum strength of surface magnetic field is $B_{*,\rm max} \sim \sqrt{G}M/r_g^2$ with r_g as the horizon radius, and the maximum magnetic flux from the object is of order $\Phi_{B,\rm max} \sim \sqrt{G}M$. Thus the mass of the object for electron acceleration is constrained as

$$M < \epsilon c^4 r_0^2/(6G^{3/2}e) \sim 660\,M_\odot(m/m_e)^{-2}, \quad (6)$$

which shows that the surface of magnetars, stellar mass black holes and intermediate mass black holes with very strong magnetic fields and fast rotation can be the candidates for electron acceleration.

Besides field and mass constraint, another necessary requirement for particle acceleration near the surface required the acceleration length should be less than the size of the object, otherwise it will fly away before being accelerated to sufficient high energy. Now the acceleration length should be $ct_{\rm acc} < r_*$, or $E/(eE_l) = \gamma E/(e\epsilon B) < r_*$, which leads to a universal upper bound of particle acceleration regardless of its mass as $E_{\rm max} \sim \epsilon e\Phi_B r_*^{-1} < \epsilon e\Phi_{B,\rm max} r_g^{-1} \simeq \epsilon e c^2/(2\sqrt{G})$, or

$$\begin{aligned} E_{\rm max} &\approx \frac{\epsilon}{2}\alpha^{1/2}E_{\rm pl}\left(\frac{\Phi_B}{\Phi_{B,\rm max}}\right)\left(\frac{r_g}{r_*}\right) \\ &\approx 3\times 10^{26}\,\mathrm{eV}\left(\frac{\epsilon}{0.5}\right)\left(\frac{\Phi_B}{\Phi_{B,\rm max}}\right)\left(\frac{r_g}{r_*}\right), \quad (7)\end{aligned}$$

where $E_{\rm pl}$ and α are the Planck energy and fine structure constant respectively, and we take $\epsilon = 0.5$ and r_g as the Schwarzschild radius. Electrons can be accelerated to maximum energy as $\alpha^{1/2}E_{\rm pl}$ ($\sim 10^{27}$ eV) only at the object horizon surface $r = r_g$ with a maximum magnetic flux and $\epsilon = 1$ around a black hole.

Relativistic MHD winds from the central object or the surrounding torus along the rotation axis are produced due to various mechanisms, and it can also accelerate electrons to a ultra high energy. Although the electromagnetic fields at large radius is much weaker than the surface's, the initial wind bulk Lorentz factor Γ helps electrons in the shell to be accelerated, as we assume the initial Γ is gained due to macrophysical energy input and as a constant as least in the timescale we concern. Taking the bulk Lorentz factor of the shell as Γ, the shell has the particle acceleration region $\delta r = \delta r'/\Gamma = r/\Gamma^2$, the light traveling time t and object variable timescale δt satisfy $r = ct = c\delta t\Gamma^2$ [14]. First of all we adopt a monopole magnetic field topology around the compact object as $B \propto r^{-2}$, the magnetic field flux is conserved from the object surface. From equation (5) in the shell comoving frame we have $B\Gamma = \Phi_B\Gamma/r^2 = \Phi_B\Gamma/(c\delta t\Gamma^2)^2 < 3\epsilon' e/(2r_0^2)$. This inequality combined with the constraint $\delta t \geq r_g/c \simeq 2GM/c^3$ gives the maximum shell Γ allowed for electron acceleration in a shell as

$$\Gamma_{\rm max} \sim \left(\frac{2\Phi_B r_0^2\epsilon'}{3ec^2\delta t^2}\right)^{1/3} \sim 9\epsilon'^{1/3}(M/M_\odot)^{-1/3}(m/m_e)^{-2/3}, \tag{8}$$

Combing equations (5) and (8) we obtain the minimum allowed magnetic field for acceleration as

$$B_{*,\rm min} \sim 3e/(2\epsilon' r_0^2\Gamma_{\rm max}) \sim 3\times10^{15}\,{\rm G}\,(m/m_e)^{8/3}\,(M/M_\odot)^{1/3} \tag{9}$$

which gives a critical radius $r_{*,c} \sim 1\times10^7$ cm $(M/M_\odot)^{1/3}(m/m_e)^{-4/3}$. This requirement can be achieved in the magnetized torus around compact objects. For $r_* \neq r_{*,c}$ or $\Phi_B \neq \Phi_{B,\rm max}$, the surface field should be always stronger than $B_{*,\rm max}$ for acceleration requirement. For more general magnetic field topology $B \propto r^{-n}$ ($n \geq 2$), the maximum ejecta Lorentz factor is required as $\Gamma < [2\Phi_B r_*^{n-2} r_0^2\epsilon'/(3ec^n\delta t^n)]^{1/(2n-1)} < [\Phi_{B,\rm max} r_0^2\epsilon_{0.5}/(3e)]^{1/(2n-1)}$, and $B_* > [3e/(2r_0^2\epsilon')]^{2n/(2n-1)}\Phi_{B,\rm max}^{-1/(2n-1)}$.

Importantly in the wind, electrons can be accelerated to a energy of a factor of Γ lower than the surface acceleration as

$$E_{\rm max}^{(\rm wind)} \approx \frac{\epsilon'\alpha^{1/2}E_{\rm pl}}{2\Gamma} > 6\times10^{25}\,{\rm eV}\,\epsilon'^{2/3}(M/M_\odot)^{1/3}(m/m_e)^{2/3} \tag{10}$$

This result is made by taking the acceleration length in the shell comoving frame $ct'_{\rm acc}$ to be less than the shell size $\delta r'$, or $E'/(e\epsilon' B') < \delta r' = r/\Gamma$. Remember that in the shell frame $E' = E\Gamma$ and $B' = \Gamma B$. We focus on the case that the radial thickness $\delta r'$ is much smaller than angular expansion $r_\perp$, which requires the shell expanded diameter angular to be larger than 8°.

Magnetars, in principle can generate magnetic fields as strong as 3×10^{17}(P/ms)$^{-1}$ G via differential rotation and penetrate through the surface by buoyancy instability around a fraction of seconds [15]. Taking the surface field $B_* = 10^{17}$ G $B_{*,17}$, $r_* = 10^6$ cm $r_{*,6}$, period P=(P/ms) ms, the surface magnetic flux is $\Phi_B \sim 10^{29}$ G cm^2 $B_{*,17}r_{*,6}^2$, and $\epsilon \sim 2\pi r_*/(cP) = 0.2r_{*,6}$ (P/ms)$^{-1}$. Electrons can be accelerated to a maximum energy as $E_{\rm max} \sim e\epsilon\Phi_B/(r_*) \sim 6\times10^{24}$ eV $B_{*,17}r_{*,6}$ (P/ms)$^{-1}$. Although the maximum accelerated electron from magnetars is two orders of magnitude lower than the highest one in the black hole case, the predicted $\sim 10^{24}$ eV electron energy is still much significantly higher than what have been considered before [10–12].

Secondary Cosmic Rays and Neutrinos. –Inverse Compton scattering is the most important classical mechanism for electron cooling besides electromagnetic radiation. The region where the electron is accelerated around the object is $r_{\rm acc} \sim c\delta t\Gamma^2 < 1.4\times10^7$ cm $(M/M_\odot)^{-2/3}(m/m_e)^{-4/3}$. On the other hand, as the scattering Klein-Nishina section decreases significantly with high electron energy, the farthest distant a electron can go with the maximum lorentz factor $\gamma_{\rm max} = E_{\rm max}/(m_ec^2)$ is $r_{\rm IC} \sim 2mc^2/[\sigma_T U_{\rm rad}\ln(2\gamma)] \gg r_{\rm acc}$. That means the electron can be accelerated before scattering with photon background. Also, we can check that e^-e^+ or e^--hadron interaction is also less important than acceleration within the electron acceleration region $r_{\rm acc}$. The electron can be accelerated to the maximum energy without significant energy loss.

Proton Acceleration and Discussion. – For protons, we use the similar method to obtain the criteria for acceleration to the maximum energy in equation (7). The mass of the object and the surface magnetic field are constrained as

$$\begin{aligned} M &< 90(m/m_p)^{-2}M_\oplus \\ B\Gamma &> 6.1\times10^{22}(m/m_p)^2\,{\rm G}, \end{aligned} \tag{11}$$

where $\Gamma < 3(M/M_\oplus)^{-1/3}(m/m_p)^{-2/3}$ is the bulk Lorentz factor of the ejected winds which are able to accelerate protons. Therefore the only possible candidate for proton acceleration to a energy level $\sim \alpha^{1/2}E_{\rm pl}$ in strong fields are earth-mass black holes, or say primordial black holes left since early Universe epoch [16]. As whether they exist is still unknown, we have not emphasized proton acceleration. The extremely strong field $B \geq 10^{22}$ G (weird, other effects)...

GR effect to accelerate particle to a Planck-scale energy [17]

Conclusions. –

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